

AGENTIC AI MINI PROJECT — CASE STUDY

Student Health and Lifestyle Mentor Agent

N. Harshitha | A. Aishwarya | D. Bharathi | K. Nandini
B.Tech Artificial Intelligence and Machine Learning | 2024–25

1. INTRODUCTION

College students frequently struggle to maintain a healthy lifestyle due to academic pressures, irregular schedules, and lack of personalized guidance. The Student Health and Lifestyle Mentor Agent is an Agentic AI-based intelligent assistant that addresses this gap by offering diet planning, fitness coaching, sleep analysis, and mental wellness support — all within a single conversational platform.

2. PROBLEM STATEMENT

Students face poor dietary habits, physical inactivity, disturbed sleep, and high stress levels. Existing health apps offer only static, generic advice without understanding individual needs. There is a clear need for an intelligent, personalized, and conversational AI-powered health assistant that adapts to each student's unique health profile and provides multi-domain guidance.

3. OBJECTIVE

- Provide personalized diet and calorie plans.
- Suggest exercise, yoga, and fitness routines.
- Analyze sleep patterns and improve quality.
- Deliver stress management and mental wellness tips.
- Calculate BMI and recommend daily water intake.
- Answer health queries using RAG from PDF documents.

4. PROPOSED SOLUTION

A multi-agent AI system using CrewAI and LangChain with four specialized agents — Diet, Fitness, Mental Wellness, and a Main Orchestrator. The system is powered by Google Gemini 2.5 Flash as the LLM and ChromaDB as the vector database for RAG. Streamlit provides an interactive web interface with conversation memory for personalized interactions.

5. TECHNOLOGIES USED

Component	Technology
Frontend	Streamlit
Backend	Python
Agent FW	LangChain
Multi-Agent	CrewAI
LLM	Gemini 2.5 Flash
Vector DB	ChromaDB
Embeddings	Google GenAI

6. TEAM MEMBERS & RESPONSIBILITIES

Member	Module
N. Harshitha	Diet & Nutrition Agent
A. Aishwarya	Fitness Coach Agent
D. Bharathi	Mental Wellness Agent
K. Nandini	Main Agent & Integration

7. EXPECTED OUTCOMES

- Personalized, evidence-based health guidance for students.
- Improved diet, fitness, sleep, and mental wellness habits.
- A scalable, real-world Agentic AI demonstration.
- Hands-on experience with CrewAI, LangChain, and Gemini.

8. CONCLUSION

This project demonstrates a practical and impactful application of Agentic AI for student health. By integrating multi-agent orchestration, RAG-based knowledge retrieval, and LLM-powered conversations, it delivers a comprehensive and intelligent health mentoring solution with strong academic and real-world value.

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ABSTRACT

The Student Health and Lifestyle Mentor Agent is a multi-agent Agentic AI system built on CrewAI, LangChain, and Google Gemini 2.5 Flash to provide personalized health support for students. Four specialized agents handle diet, fitness, mental wellness, and system orchestration. A RAG pipeline backed by ChromaDB retrieves knowledge from curated health PDFs, while Streamlit delivers an interactive conversational interface with persistent memory.

EXISTING VS. PROPOSED SYSTEM

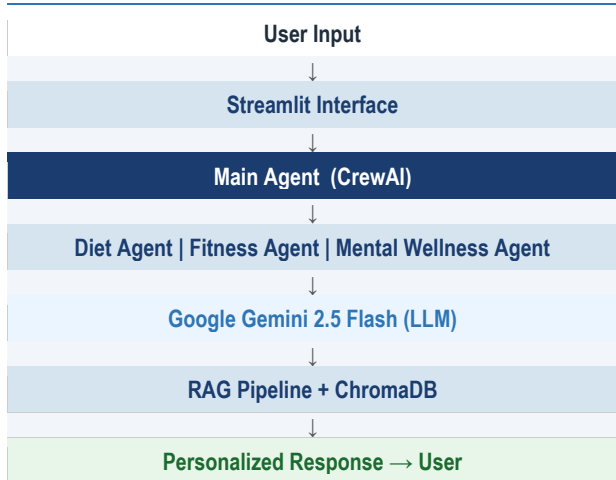
Existing System Limitations

- Generic recommendations, no personalization.
- No conversational or multi-turn AI capability.
- Separate apps needed for diet, fitness, mental health.
- No RAG from domain-specific health documents.

Proposed System Advantages

- Multi-agent AI with CrewAI for specialized reasoning.
- Personalized responses via conversation memory.
- Single platform for all health domains.
- RAG from curated PDFs via ChromaDB.

SYSTEM ARCHITECTURE



WORKFLOW

- Student submits a health query via Streamlit.
- Main Agent (CrewAI) analyzes intent and delegates task.
- Specialized agent uses LangChain tools + Gemini LLM.
- RAG pipeline fetches relevant content from ChromaDB.
- Agents synthesize a final personalized response.
- Response displayed; conversation saved to memory.

MODULE DESCRIPTIONS

Diet & Nutrition Agent — N. Harshitha

Handles BMI calculation, calorie recommendations, and personalized meal planning. Maintains a nutrition knowledge base from curated health PDFs for RAG. Deliverables: diet_tool.py, Diet PDF Collection, RAG Data Preparation.

Fitness Coach Agent — A. Aishwarya

Provides exercise recommendations, workout plans, and yoga suggestions based on the student's fitness level and goals. Deliverables: exercise_tool.py, Fitness Knowledge Base.

Mental Wellness Agent — D. Bharathi

Delivers stress management techniques, sleep pattern analysis, and motivational support. Analyzes reported sleep habits and suggests actionable improvements. Deliverables: stress_tool.py, sleep_tool.py.

Main Agent & Integration — K. Nandini

Orchestrates all agents via CrewAI, manages conversation memory, and deploys the Streamlit UI. Responsible for end-to-end integration and testing. Deliverables: app.py, agents.py, memory.py.

FUNCTIONAL REQUIREMENTS

FR	Requirement
FR1	Calculate BMI from height and weight.
FR2	Recommend daily water intake.
FR3	Provide personalized diet plans.
FR4	Suggest exercise and workout plans.
FR5	Analyze sleep and provide tips.
FR6	Stress management and mental wellness.
FR7	RAG-based health Q&A from PDFs.
FR8	Conversation memory for personalization.
FR9	Multi-agent support via CrewAI.
FR10	Interactive web UI via Streamlit.

FUTURE ENHANCEMENTS

- Wearable device integration for real-time data.
- Voice interaction (speech-to-text / TTS).
- Medical report upload and analysis.
- Multilingual support for regional languages.
- Mobile app for Android and iOS.
- Visual health dashboard with trend charts.

CONCLUSION

The Student Health and Lifestyle Mentor Agent is a comprehensive demonstration of Agentic AI applied to student well-being. Through multi-agent orchestration, RAG, and LLM integration, it delivers intelligent, personalized health guidance. The modular design, strong technology stack, and clear team contributions make this a valuable B.Tech AIML mini project with real-world applicability.